H

RAFFLES INSTITUTION
2018 Year 6 Preliminary Examination

Higher 1

CANDIDATE NAME	

CIVICS GROUP	1	8	S	0	3		INDEX NUMBER							

BIOLOGY **8876/02**

Paper 2 Structured and Free-response Questions **13th September 2018**

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES

Section A
Answer **all** questions in the spaces provided on the Question Paper.6
Section B
Answer any **one** question in the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, **hand in your essay question SEPARATELY**.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
SECTION A	
1	**/12**
2	**/9**
3	**/11**
4	**/7**
5	**/6**
SECTION B	
6 or 7	**/15**
Total	**/60**

This document consists of **12** printed pages.

Raffles Institution
Internal Examination

[Turn over

For Examiner's Use

SECTION A
Answer **all** questions.

1 Phosphofructokinase **(**PFK) is an allosteric enzyme made of 4 subunits and controlled by many activators and inhibitors which regulates glycolysis. PFK phosphorylates fructose-6-phosphate to form fructose-1,6-bisphosphate.This enables the cell to increase or decrease the rate of glycolysis in response to the cell's energy requirements. For example, a high ratio of ATP to ADP will inhibit PFK and glycolysis.

Fig 1.1 shows the effect of low ATP on Phosphofructokinase (PFK) activity.

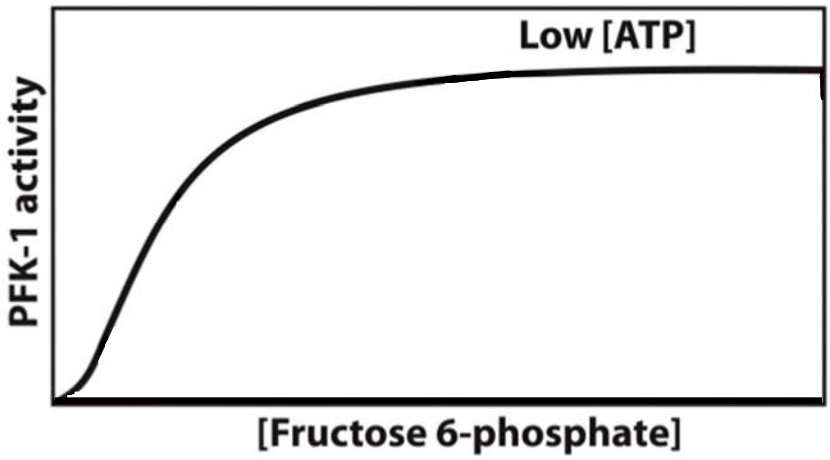

Fig 1.1

(a) **(i)** On Fig. 1.1, draw a graph to show the effect of high ATP on PFK activity. [1]

(ii) Explain the graph you have drawn.

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(iii) Name a molecule that will act as an allosteric activator of PFK.

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(b) Fig. 1.2 shows a PFK molecule.

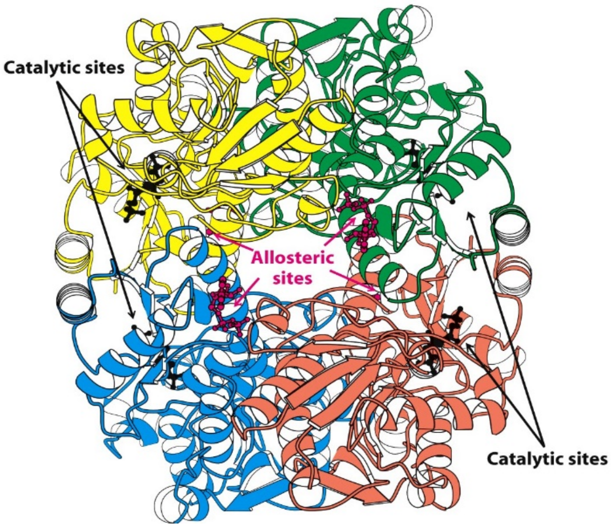

Fig 1.2

(i) Describe the structure of PFK.

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(ii) Explain how a change in pH may affect PFK activity.

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[Total : 12]

2 During meiosis, the cell divides to produce gametes.

(a) Discuss the role of centromere in the production of gametes.

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(b) Most human traits as well as medical conditions are under genetic influence.

A gene is found to control a rare disease, Wiskott–Aldrich syndrome (WAS) which is characterised by eczema, low platelet count and immune deficiency.

Another gene that controls hair type has 2 alleles. 1 allele results in straight hair, another codes for curly hair. Presence of both results in wavy hair.

In **couple 1**, a normal female with straight hair married a wavy-haired male suffering from WAS. Predicted phenotypic ratio of their offspring is as follows:

	WAS	Hair
Female	All normal	1 wavy : 1 straight
Male	All normal	1 wavy : 1 straight

For **couple 2**, a wavy-haired female suffering from WAS married a wavy-haired normal man. Predicted phenotypic ratio of their offspring is as follows:

	WAS	Hair
Female	All normal	1 curly : 2 wavy : 1 straight
male	All affected	1 curly : 2 wavy : 1 straight

(i) What is the mode of inheritance of the WAS disease?

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(ii) Use suitable symbols to represent the alleles of the gene controlling WAS disease.

…………………………………………………………………………………………[1]

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(iii) Use a genetic diagram to explain the results of **couple 2**.

[4]

[Total : 9]

3 **Fig. 3.1** shows the results of experiments investigating the effect of different light intensities on the rate of photosynthesis of cucumber plants measured as mm^3 CO_2 uptake per cm^2 leaf area per hour. The experiments were carried out at different temperatures and carbon dioxide concentrations.

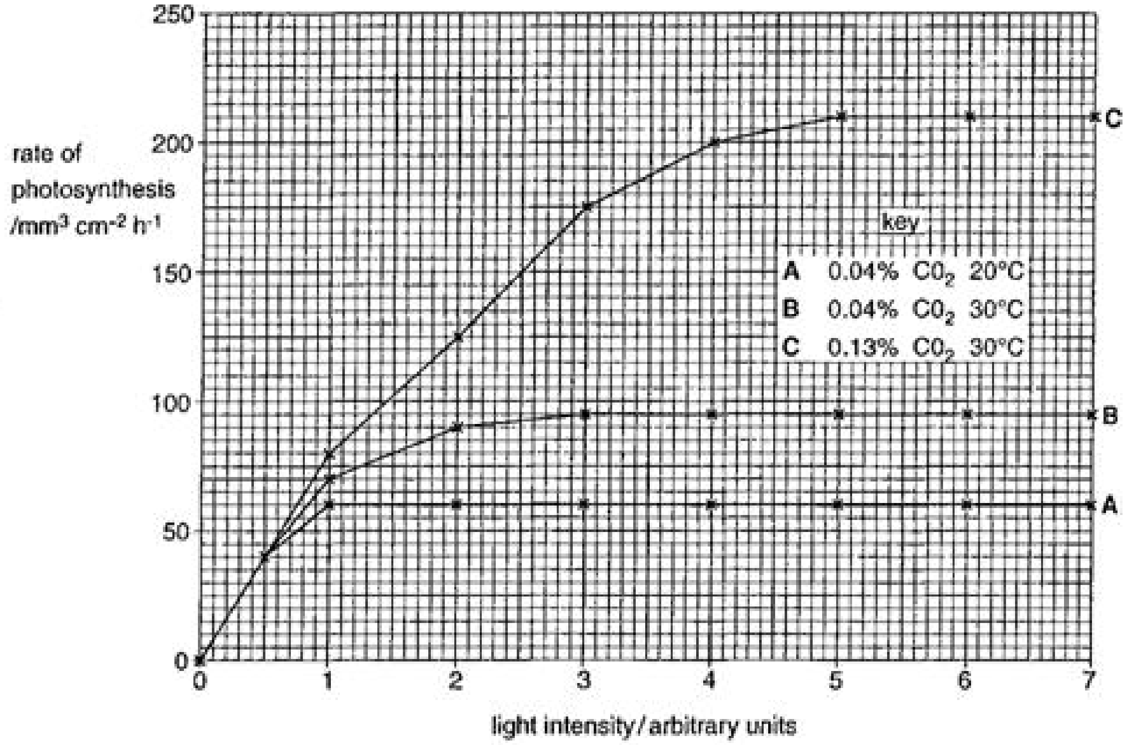

Fig 3.1

[Turn over

For Examiner's Use

(a) **(i)** With reference to Fig. 3.1, state the best conditions for the growth of cucumber.

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(ii) With reference to Fig. 3.1, explain reasons for the difference between curves **B** and **C**.

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(b) Fig. 3.2 shows the absorption spectra of various leaf pigments and the action spectrum for photosynthesis.

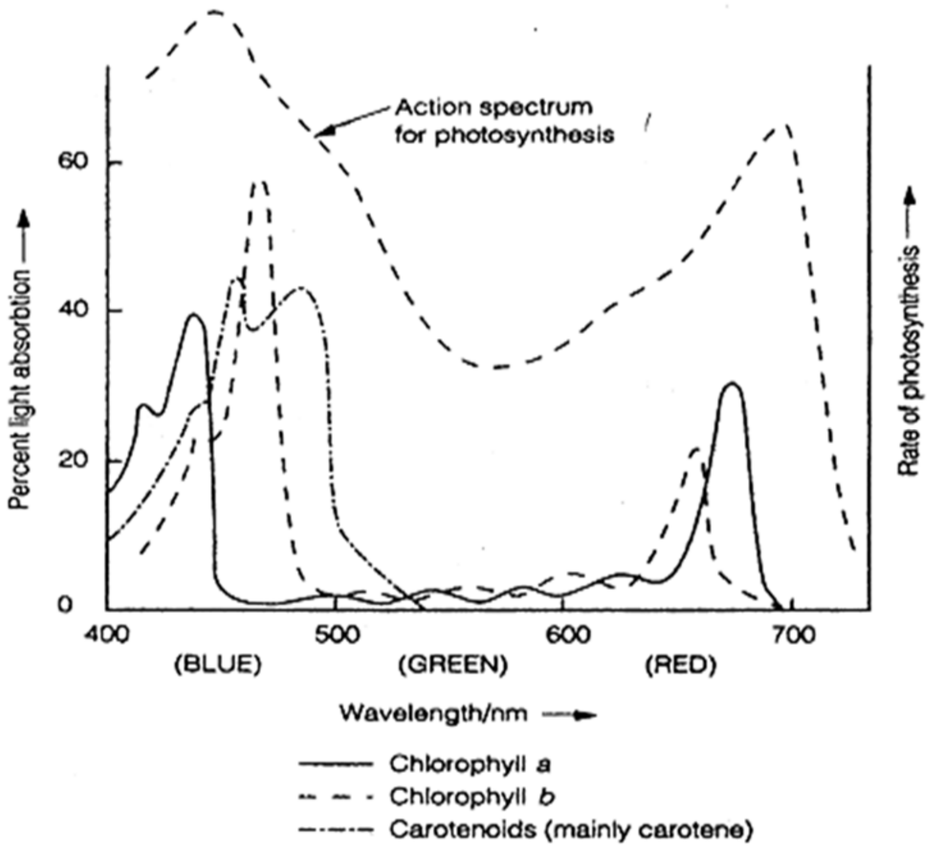

Fig. 3.2

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With reference to Fig. 3.2,

(i) explain why most plants are characteristically green.

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(ii) explain the shape of the action spectrum.

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(c) In conducting experiments to obtain the action spectrum for photosynthesis, explain why the use of organic solvents such as acetone or ether is necessary to extract photosynthetic pigments.

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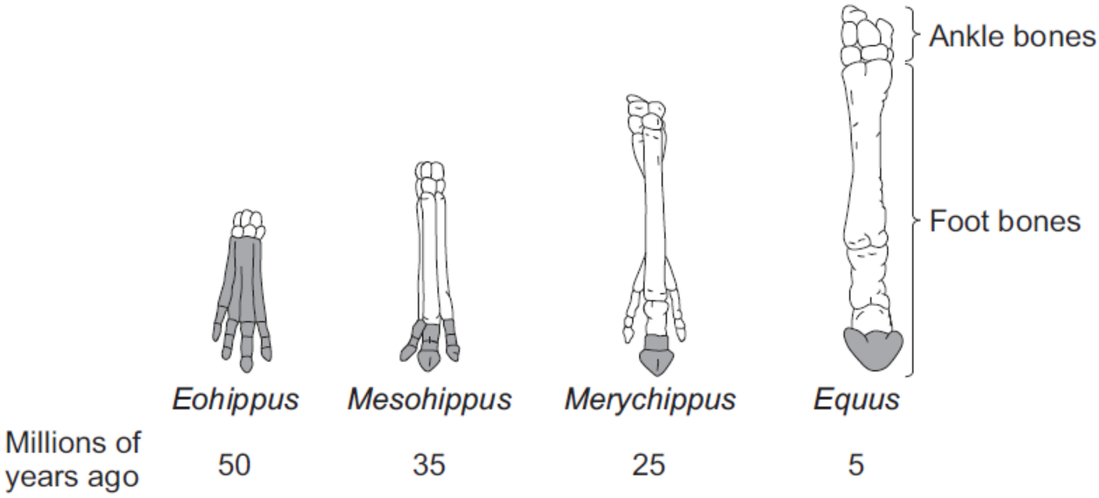
[Total: 11]

For Examiner's Use

4 Fig. 4.1 shows changes in the foot bones of four ancestors of modern horses over the past 50 million years.



Key: The shaded bones are the bones which touched the ground.

Fig. 4.1

(a) Describe two changes in the foot bones of horses that have taken place over the past 50 million years.

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(b) *Eohippus* lived in swampy areas with soft mud.
Since then, the ground in the habitat has become drier and harder.
All of the horse ancestors were preyed upon by other animals.

(i) Explain **one** advantage to *Eohippus* of the arrangement of bones in its feet.

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For Examiner's Use

(ii) Based on the information given, explain how the changes in the arrangement of the foot bones of horses support Darwin's theory of evolution by natural selection.

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[Total: 7]

5 Sea ice is an integral part of the Arctic Ocean. The extent of area covered by Arctic sea ice is an important indicator of changes in global climate because warmer air and water temperatures are reducing the amount of sea ice present.

Fig. 5.1 shows Arctic sea ice extent for the months of September and March of each year from 1979 through 2016. September and March are when the minimum and maximum extent typically occur each year.

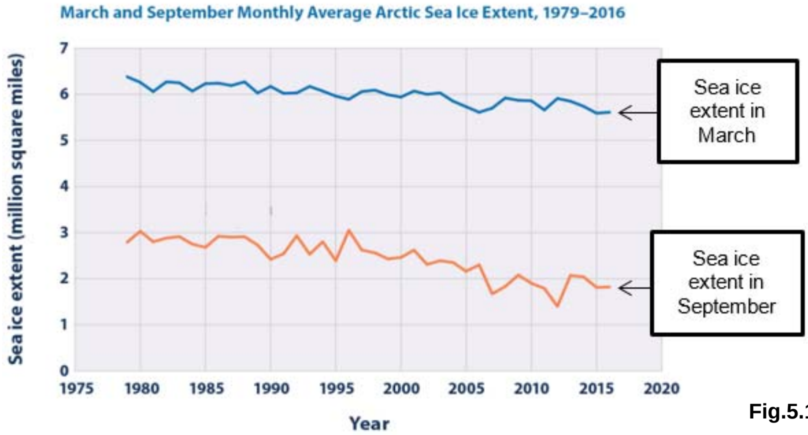

Fig.5.1

Fig. 5.2 shows the start and end of the sea ice melt season.

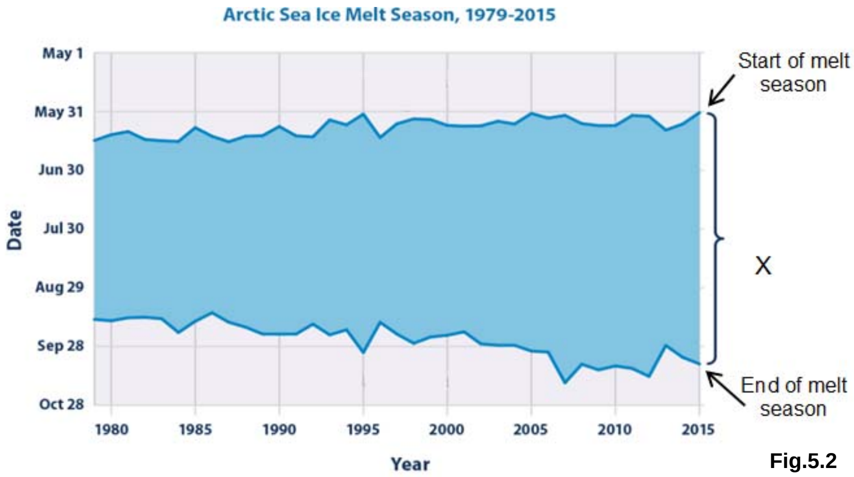

Fig.5.2

Fig. 5.3 shows the distribution of Arctic sea ice extent by age group during the week in September with the smallest extent of ice for each year.

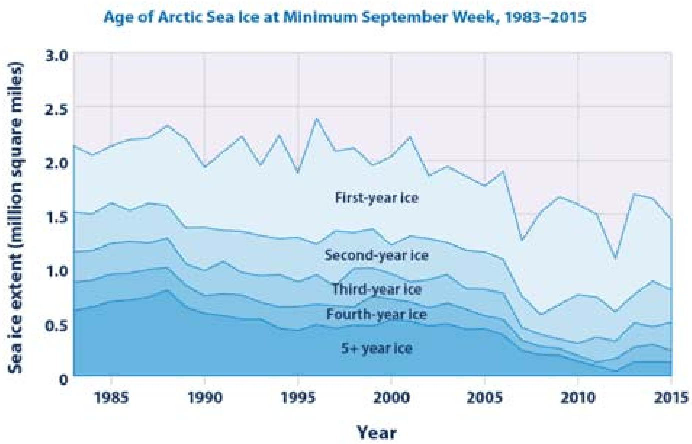

Fig.5.3

(a) With reference to Fig. 5.1, suggest why the sea ice extent in March is different from that in

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September.

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(b) With reference to Fig. 5.2,

(i) identify what **X** denotes.

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(i) Describe how **X** changes from 1979 to 2015.

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(ii) Explain how this is linked to the changes in sea ice extent as shown in Fig. 5.1 and Fig. 5.3.

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[Total: 6]

Section C

Answer one question in this section.

Write your answers on the separate writing paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

6 (a) Account for the importance of hydrogen bonds in biological organisms. [7]

(b) Describe the main features of replication of DNA. [8]

[Total: 15]

OR

7 (a) Discuss the diverse roles of membranes in a cell. [7]

(b) Describe the major properties of enzymes and discuss their mode of action. [8]

[Total: 15]

End of Paper